2025 14th International Conference on Modern Circuits and Systems Technologies (MOCAST) | 979-8-3315-3914-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/MOCAST65744.2025.11083958

People Counting and Positioning Using Low-Resolution Infrared Images for Resource-Constrained Edge Devices

Alptekin Vardar*†, Li Zhang*†, Marc Lupp‡, Alexander Rehmer‡,
Nandakishor Yadav†, and Thomas Kämpfe†§
*These authors contributed equally to this work.
†Fraunhofer IPMS, Germany ‡Heimann Sensor GmbH, Germany §TU Braunschweig, Germany
Email: alptekin.vardar@ipms.fraunhofer.de, thomas.kaempfe@ipms.fraunhofer.de

***Abstract*—This paper presents a people counting and indoor positioning system using low-resolution infrared video from a thermopile sensor, optimized for deployment on resource-constrained edge devices. A lightweight YOLOv3-tiny network is trained on a custom thermal dataset and enhanced through quantization-aware training and structured pruning. Pre-processing techniques such as Otsu thresholding and morphological filtering are used to suppress noise and reduce image size by over 80 percent with minimal accuracy loss. The final model—quantized to 4 bits and pruned by 50 percent—achieves 57.71 precent mAP at 0.5 while significantly lowering memory and compute demands. Results validate the system's suitability for privacy-preserving, low-power applications in smart environments.**

***Index Terms*—Infrared imaging, people counting, indoor positioning, edge inference, YOLOv3-tiny, quantization, pruning**

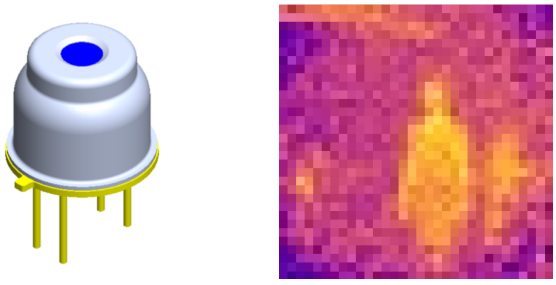
I. Introduction

Real-time people counting and indoor positioning are critical for diverse applications including building automation, occupancy monitoring, energy management, and security surveillance [1]. Conventional approaches, which typically rely on high-resolution RGB or depth sensors, require considerable computational resources, thereby limiting their deployment on resource-constrained edge devices [2]. Consequently, there exists a compelling need for solutions that strike an optimal balance between accuracy, efficiency, privacy, and computational feasibility [3].

Infrared (IR) imaging emerges as a promising alternative, offering robust performance under challenging environmental conditions such as low-light or visually obstructed scenarios [4]. Moreover, the inherent characteristics of IR imaging—capturing thermal signatures rather than detailed visual features—ameliorate privacy concerns in sensitive environments. Nonetheless, deploying IR-based detection systems on edge devices presents substantial challenges. Limited processing power and stringent energy constraints necessitate the development of lightweight, computationally efficient models capable of real-time performance without compromising detection accuracy [5].

This work addresses these constraints by utilizing a compact thermopile-based infrared sensor developed by Heimann Sensors to capture low-resolution thermal video in a controlled indoor setting. Using this proprietary dataset, we implement a lightweight detection pipeline centered on YOLOv3-tiny, enhanced through quantization-aware training and structured pruning to reduce computational load. In parallel, a series of targeted pre-processing steps—background subtraction, thresholding, and morphological filtering—are applied to suppress noise and reduce data throughput before inference. Combined, these approaches enable efficient, real-time people detection and positioning on constrained edge hardware without compromising accuracy.

A. Infrared Sensor



(a) HTPA32x32L1.9 (b) Exemplary sensor image

Fig. 1: a) Thermopile array sensor employed in this application; b) An exemplary frame from the dataset recorded with this sensor.

The IR sensor used for data acquisition in this study is a thermopile array. A thermopile converts thermal energy into electrical energy using the Seebeck effect. It consists of multiple thermocouples connected electrically

979-8-3315-3914-6/25/$31.00 ©2025 IEEE
Authorized licensed use limited to: University of Limerick. Downloaded on October 12,2026 at 11:30:03 UTC from IEEE Xplore. Restrictions apply.

in series and thermally in parallel. When exposed to a temperature gradient, the junctions of different conductive materials generate a voltage proportional to the difference between the sensor's own temperature and that of the measured object. By arranging multiple thermopiles in an array, an imaging infrared sensor is created, making it suitable for thermographic applications such as person detection and tracking. The specific thermopile array used in this study has a resolution of 32×32 pixels and a field of view of $100°$, both horizontally and vertically. A CAD model of the thermopile array is shown in Fig.1a.

II. Methodology

A. Dataset

The dataset consists of multiple short clips (1–2 minutes each) recorded in an open-plan office-like environment. The sensor was mounted at a height of 1.50 meters above the ground, with an elevation angle of $0°$ (parallel to the ground plane), resulting in an eye-level perspective. An exemplary frame from the recorded dataset is shown in Fig. 1b. The clips feature between one and four individuals at a time, primarily in a standing position. Approximately 180 frames were labeled, and the resulting dataset was split into training (80%), validation (10%), and testing (10%) subsets. Although the labeled dataset was relatively small, extensive data augmentation and the compact YOLOv3-tiny architecture helped mitigate overfitting and support generalization. Data augmentation techniques included horizontal flipping, shift-scale-rotate transformations, brightness adjustments, RGB shifting, and Gaussian blurring [6].

B. Pre-processing Techniques

Pre-processing plays a critical role in improving detection performance and computational efficiency when working with low-resolution infrared data. Given the sensor's limited resolution and inherent thermal noise, tailored techniques are required to enhance relevant features while suppressing background artifacts. This study evaluates several techniques designed to improve signal clarity and reduce redundant information:

- **Background Subtraction**: Separates moving objects from a static scene by subtracting a reference frame [7], highlighting human presence while suppressing stationary heat sources.
- **Grayscale Conversion**: Converts multi-channel thermal images into a single-channel format to reduce complexity while retaining intensity information [8].
- **Thresholding (Otsu's Method)**: Applies a data-driven threshold to segment foreground and background based on intensity variance [9], particularly effective in thermally consistent environments. Though originally designed for grayscale images, Otsu's method is effective in thermal imaging scenarios due to the typically bimodal distribution of pixel intensities representing human heat signatures and ambient background.
- **Morphological Operations**: Uses erosion and dilation to refine segmented regions [10]. Erosion removes noise; dilation restores object shape.

Each technique involves trade-offs between efficiency, robustness, and adaptability. Background subtraction excels in static scenes but falters with dynamic changes. Grayscale conversion is lightweight but may obscure subtle thermal details. Thresholding provides sharp segmentation yet is sensitive to ambient variation. Morphological operations enhance structure but add computation. In this work, we combine these steps to reduce data redundancy while preserving key features and detection performance.

C. YOLOv3-tiny Network

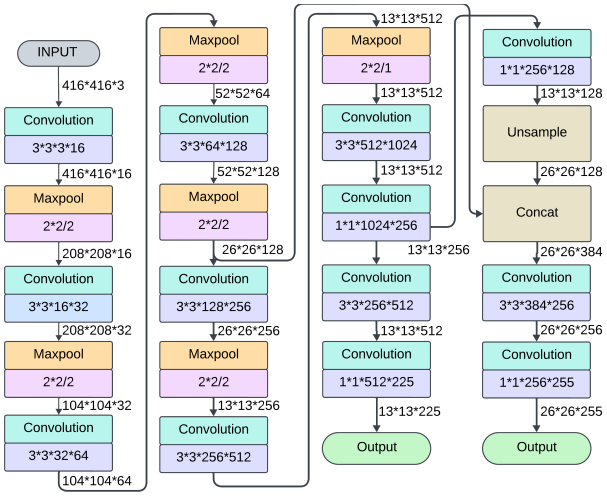


Fig. 2: YOLOv3-tiny architecture used in this work. The network consists of 13 convolutional layers, with detection performed at two different feature map scales to support multi-size object detection.

YOLOv3-tiny [11] is a lightweight variant of the YOLOv3 object detection framework, designed to deliver real-time performance with reduced computational complexity. Its architecture comprises 13 convolutional layers and approximately 8.7 million parameters, derived from a simplified version of the Darknet-53 backbone. This compact design makes it well-suited for edge deployment, particularly in scenarios with constrained resources. Unlike two-stage detectors (e.g., Faster R-CNN [12]) that rely on region proposals, YOLOv3-tiny employs a single-stage architecture that directly predicts object locations and class probabilities in one forward pass. This design is advantageous for infrared data,

Authorized licensed use limited to: University of Limerick. Downloaded on October 12,2026 at 11:30:03 UTC from IEEE Xplore. Restrictions apply.

where low contrast and inherent noise can hinder region-based methods.

The network divides each input frame into an $S \times S$ grid. Each grid cell predicts bounding box coordinates, objectness scores, and class probabilities using predefined anchor boxes of varying aspect ratios, enabling robust multi-scale detection. Predictions are filtered using an Intersection over Union (IoU) threshold to retain only high-overlap detections, followed by confidence score filtering and Non-Maximum Suppression (NMS) to eliminate redundant boxes. A schematic overview of the YOLOv3-tiny architecture is shown in Fig. 2.

D. Evaluation Metric

To assess the performance of the proposed detection pipeline, we employ Mean Average Precision (mAP), a standard metric in object detection tasks [13]. mAP captures both localization and classification performance by averaging the precision-recall characteristics across all classes.

Intersection over Union (IoU) measures the overlap between the predicted bounding box B_p and the ground truth B_g:

$$IoU = \frac{|B_p \cap B_g|}{|B_p \cup B_g|} \tag{1}$$

A prediction is considered correct if its IoU exceeds a predefined threshold, typically set at 0.5 (denoted as mAP@0.5).

Precision and **Recall** are defined as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \tag{2}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively.

Average Precision (AP) is computed as the area under the precision-recall (PR) curve:

$$AP = \int_0^1 P(R)\, dR \tag{3}$$

Mean Average Precision (mAP) is then defined as the mean of AP over all classes:

$$mAP = \frac{1}{N} \sum_{i=1}^{N} AP_i \tag{4}$$

where N is the number of object classes.

In this work, we report mAP at an IoU threshold of 0.5 (mAP@0.5), which serves as a robust indicator of both detection accuracy and localization quality.

E. Hardware-Aware Training

To ensure real-time inference on resource-constrained edge devices, we apply two complementary optimization techniques: quantization-aware training (QAT) and structured pruning. These approaches reduce model complexity while preserving accuracy.

1) Quantization-Aware Training

QAT integrates quantization effects during training, allowing the model to adapt to reduced numerical precision and maintain performance [14]. Using QKeras [15], we quantize weights and activations to fixed-point representations (e.g., 8-bit, 4-bit, 2-bit). Batch normalization folding is also applied to reduce inference latency. The quantization function is defined as:

$$Q(x) = \text{clip}\left(\text{round}\left(\frac{x}{s}\right) \cdot s, x_{\min}, x_{\max}\right) \tag{5}$$

2) Structured Pruning

Structured pruning removes entire channels from convolutional layers, enabling efficient execution on standard hardware [16]. Following the Network Slimming approach [17], we impose ℓ_1 regularization on batch normalization scaling factors γ to promote sparsity:

$$L = \sum_{(x,y)} l(f(x, W), y) + \lambda \sum_{\gamma \in \Gamma} |\gamma| \tag{6}$$

Channels with low γ values are pruned, followed by fine-tuning to recover performance. We evaluate multiple pruning levels to balance efficiency and model capacity.

III. Results

A. Impact of Image Compression

We evaluated how pre-processing affects detection accuracy and image size. The formats compared include raw infrared frames, background-subtracted grayscale (BS_Gray), Otsu thresholding (Gray_OTSU), and Otsu with morphological filtering (Gray_OTSU_Morph). The results are summarized in Fig. 3.

While background-subtracted grayscale achieves the highest accuracy, the Gray_OTSU and Gray_OTSU_Morph formats achieve over 80% reduction in image size with only moderate loss in detection performance. This substantial compression makes them highly suitable for edge deployments where memory and data throughput are critical. Given this trade-off, the Gray_OTSU_Morph variant was selected as the default input format for all subsequent quantization and pruning experiments.

B. Quantization, Pruning, and Deployment Efficiency

To reduce the memory footprint and computational load for edge inference, we applied different combinations of quantization and structured pruning to the

Authorized licensed use limited to: University of Limerick. Downloaded on October 12,2026 at 11:30:03 UTC from IEEE Xplore. Restrictions apply.

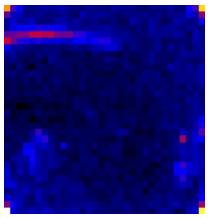
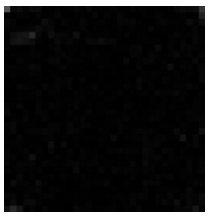
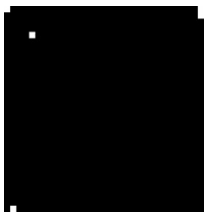
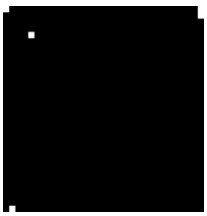
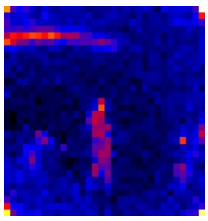
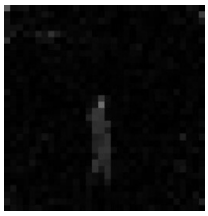
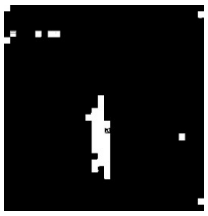
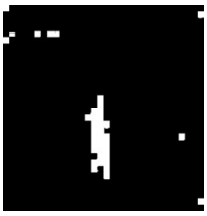
	Raw	BS_Gray	Gray_OTSU	Gray_OTSU_Morph
BG Image				
Inference Output				
Image Size	25 kB	24.2 kB	3.88 kB	3.88 kB
mAP@0.5	69.70%	72.22%	59.05%	62.94%

Fig. 3: Visual and quantitative comparison of pre-processed infrared input formats. The Gray_OTSU_Morph configuration offers a strong trade-off between compression and detection performance.

YOLOv3-tiny model and evaluated their impact on accuracy and efficiency.

Quantization reduces the numerical precision of weights and activations, significantly lowering both model size and compute demand. As shown in Fig. 4, decreasing bit-width directly reduces MAC operations and memory usage—key factors for power consumption and latency in embedded devices. Among all configurations, 4-bit quantization offers the best trade-off between detection performance and efficiency.

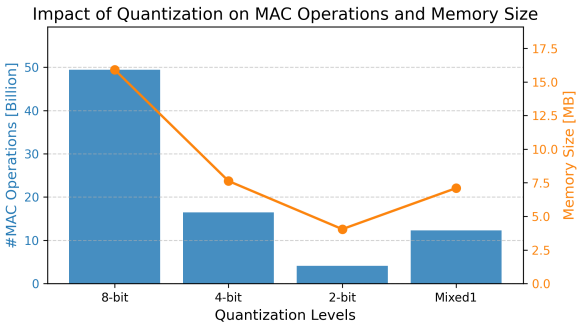


Fig. 4: MAC operations and memory usage for various quantization schemes. Lower bit-widths yield substantial energy and latency savings.

We also evaluated a mixed-precision scheme [18], where earlier layers use higher precision and later layers are more aggressively quantized. This approach achieves competitive results and shows promise, but requires further optimization to ensure stable performance across varying compression settings.

Structured pruning reduces model size by removing entire convolutional channels. Fig. 5 shows that pruning 30–50% of the network significantly lowers parameter count while preserving detection accuracy. Although 50% pruning delivers the best balance, 80% pruning remains a viable option for applications with strict memory or compute constraints.

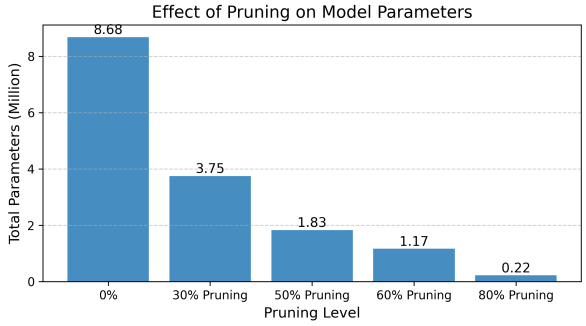


Fig. 5: Total model parameters across pruning levels. 50% pruning achieves strong compression while maintaining performance.

The combined effects of quantization and pruning are summarized in Fig. 6. The best result is achieved with 4-bit quantization and 50% pruning, reaching 57.71% mAP@0.5. This configuration provides the most favorable balance between accuracy and efficiency, supporting

Authorized licensed use limited to: University of Limerick. Downloaded on October 12,2026 at 11:30:03 UTC from IEEE Xplore. Restrictions apply.

real-time deployment on resource-limited edge devices.

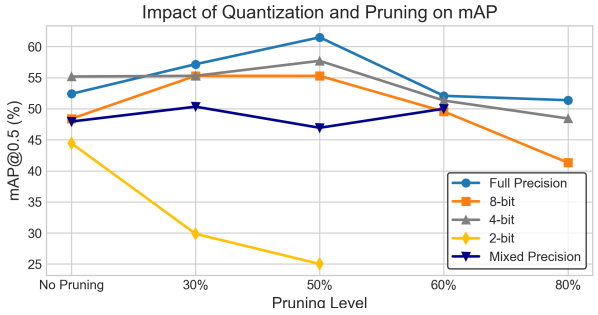


Fig. 6: mAP@0.5 for different quantization and pruning combinations. 4-bit with 50% pruning offers the best performance-efficiency trade-off.

IV. Conclusion

This work presents a lightweight, infrared-based people counting and positioning system designed for real-time deployment on resource-constrained edge devices. Leveraging low-resolution thermopile imagery, a preprocessed input pipeline, and an optimized YOLOv3-tiny architecture, the system achieves efficient inference with competitive detection performance.

Through systematic evaluation of quantization-aware training and structured pruning, we demonstrate that a 4-bit quantized model with Otsu-based binary inputs, 50% pruning offers the most effective balance between accuracy and hardware efficiency, achieving 57.71% mAP@0.5 while significantly reducing model size, memory footprint, and computational cost.

These results validate the feasibility of accurate thermal detection under extreme resource constraints, and highlight key trade-offs for future low-power vision systems. Further exploration of mixed-precision strategies and adaptive pruning could offer additional gains for ultra-constrained platforms.

Acknowledgments

This work was funded by the SensIR project. We thank the project partners for their support and collaboration throughout the development and evaluation phases.

References

[1] M. R. Holla, D. Suma, and M. D. Holla, "Optimizing accuracy and efficiency in real-time people counting with cascaded object detection," *International Journal of Information Technology*, 2024. [Online]. Available: https://doi.org/10.1007/s41870-024-02153-w

[2] C. Xie, F. Daghero, Y. Chen, M. Castellano, L. Gandolfi, A. Calimera, E. Macii, M. Poncino, and D. Jahier Pagliari, "Efficient deep learning models for privacy-preserving people counting on low-resolution infrared arrays," *IEEE Internet of Things Journal*, vol. 10, no. 15, p. 13895–13907, Aug. 2023. [Online]. Available: http://dx.doi.org/10.1109/JIOT.2023.3263290

[3] M. Bouazizi, C. Ye, and T. Ohtsuki, "Low-resolution infrared array sensor for counting and localizing people indoors: When low end technology meets cutting edge deep learning techniques," *Information*, vol. 13, no. 3, 2022. [Online]. Available: https://www.mdpi.com/2078-2489/13/3/132

[4] N. Lintvedt, "Thermal imaging in robotics as a privacy-enhancing or privacy-invasive measure? misconceptions of privacy when using thermal cameras in robots," *Digital Society*, vol. 2, no. 3, p. 33, 2023. [Online]. Available: https://doi.org/10.1007/s44206-023-00060-4

[5] M. Risso, C. Xie, F. Daghero, A. Burrello, S. Mollaei, M. Castellano, E. Macii, M. Poncino, and D. J. Pagliari, "Hw-sw optimization of dnns for privacy-preserving people counting on low-resolution infrared arrays," 2024. [Online]. Available: https://arxiv.org/abs/2402.01226

[6] C. Shorten and T. M. Khoshgoftaar, "A survey on image data augmentation for deep learning," *Journal of Big Data*, vol. 6, no. 1, pp. 1–48, 2019.

[7] O. Barnich and M. Van Droogenbroeck, "Vibe: A universal background subtraction algorithm for video sequences," *IEEE Transactions on Image Processing*, vol. 20, no. 6, pp. 1709–1724, 2011.

[8] C. Saravanan, "Color image to grayscale image conversion," in *2010 Second International Conference on Computer Engineering and Applications*, vol. 2, 2010, pp. 196–199.

[9] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.

[10] R. M. Haralick, S. R. Sternberg, and X. Zhuang, "Image analysis using mathematical morphology," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. PAMI-9, no. 4, pp. 532–550, 1987.

[11] J. Redmon and A. Farhadi, "Yolov3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.

[12] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *Advances in neural information processing systems*, vol. 28, 2015.

[13] R. Padilla, W. L. Passos, T. L. B. Dias, S. L. Netto, and E. A. B. da Silva, "A comparative analysis of object detection metrics with a companion open-source toolkit," *Electronics*, vol. 10, no. 3, 2021. [Online]. Available: https://www.mdpi.com/2079-9292/10/3/279

[14] A. Gholami, S. Kim, Z. Dong, Z. Yao, M. W. Mahoney, and K. Keutzer, "A survey of quantization methods for efficient neural network inference," 2021. [Online]. Available: https://arxiv.org/abs/2103.13630

[15] C. N. Coelho, A. Kuusela, S. Li, H. Zhuang, J. Ngadiuba, T. K. Aarrestad, V. Loncar, M. Pierini, A. A. Pol, and S. Summers, "Automatic heterogeneous quantization of deep neural networks for low-latency inference on the edge for particle detectors," *Nature Machine Intelligence*, vol. 3, no. 8, pp. 675–686, Aug. 2021. [Online]. Available: https://doi.org/10.1038/s42256-021-00356-5

[16] S. Han, J. Pool, J. Tran, and W. Dally, "Learning both weights and connections for efficient neural network," *Advances in neural information processing systems*, vol. 28, 2015.

[17] Z. Liu, J. Li, Z. Shen, G. Huang, S. Yan, and C. Zhang, "Learning efficient convolutional networks through network slimming," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, Oct 2017.

[18] A. Vardar, L. Zhang, S. Hu, S. B. Jain, S. Mojumder, N. Laleni, A. Shrivastava, S. De, and T. Kämpfe, "Layer sensitivity aware cnn quantization for resource constrained edge devices," in *2022 9th International Conference on Soft Computing Machine Intelligence (ISCMI)*, 2022, pp. 26–30.

Authorized licensed use limited to: University of Limerick. Downloaded on October 12,2026 at 11:30:03 UTC from IEEE Xplore. Restrictions apply.